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Connector

Abstract

In the present invention, a cover closes an opening in a housing holding a plurality of terminals. Each of the terminals has a connection part connecting to a mate terminal. The cover has a first surface, a second surface located at a rear side of the first surface including an exposed surface exposed to the inner side of the housing through the opening, a first reinforcement rib projecting from the first surface, and a second reinforcement rib projecting from the second surface. The direction in which the plurality of connection parts are arranged is defined as a first direction. The direction that intersects the first direction and that is oriented from the basal end toward the distal end of the connection parts is defined as a second direction. The first reinforcement rib extends along the first direction. The second reinforcement rib extends along the second direction between adjacent connection parts.

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Background/Summary

BACKGROUND

Field of the Disclosure

(1) The present disclosure relates to a connector.

Related Art

(2) In the prior art, an electrical device mounted on a vehicle is accommodated in a case. An

example of such an electrical device is electrically connected to another electrical device by a connector that is electrically connected to a terminal mount fixed to the case. The terminal mount is disposed in a mount hole of the case. The mount hole connects the inside and the outside of the case.

(3) Japanese Laid-Open Patent Publication No. 2017-92418 discloses an example of a connector connected to a terminal mount. The connector includes a connector housing including an opening, terminals held by the connector housing, and a cover closing the opening.

(4) When electrically connecting the connector to the terminal mount, first, the connector housing is disposed on the terminal mount. At this time, the cover is not coupled to the connector housing, and the opening is not closed by the cover. Subsequently, a tool is inserted into the connector housing from the opening, and the terminals are electrically connected to respective mate terminals mounted on the terminal mount with the tool. Then, the opening is closed by the cover.

(5) The temperature inside the case may be increased to a high temperature by heat generated by the electrical device and the like. This increases the pressure of the case, which increases the pressure applied to the cover from the inside of the case. Changes in the pressure may deform the cover. Deformation of the cover may form a gap between the cover and the connector housing. When a gap is formed between the cover and the connector housing, a liquid such as water may enter the inside of the connector through the gap.

(6) It is an objective of the present disclosure to provide a connector that restricts deformation of a cover.

SUMMARY

(7) A connector according to the present disclosure includes a connector housing including an opening, a cover closing the opening, and terminals held by the connector housing. Each of the terminals includes a connection part configured to be electrically connected to one of a plurality of mate terminals, the mate terminals differing from one another. The connection parts are arranged in a line in the connector housing. The cover includes a first surface exposed outside the connector housing and a second surface located at a rear side of the first surface. The second surface includes an exposed surface exposed from the opening to an inner side of the connector housing. The cover further includes at least one first reinforcement rib projecting from the first surface and at least one second reinforcement rib projecting from the second surface. When a first direction refers to a direction in which the connection parts are arranged, a second direction refers to a direction intersecting the first direction and extending from a basal end to a distal end of the connection parts, and a third direction refers to a direction orthogonal to the first direction and the second direction, the first reinforcement rib extends in the first direction, and the second reinforcement rib extends in the second direction between adjacent ones of the connection parts as viewed in the third direction.

(8) The connector according to the present disclosure restricts deformation of the cover.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view showing an embodiment of a connector.

(2) FIG. 2 is an exploded perspective view of the connector shown in FIG. 1.

(3) FIG. 3 is a plan view of a cover in the embodiment shown in FIG. 1.

(4) FIG. 4 is a side view of the cover shown in FIG. 3.

(5) FIG. 5 is a lower view of the cover shown in FIG. 3.

(6) FIG. 6 is a perspective view of the cover shown in FIG. 3.

(7) FIG. 7 is a cross-sectional view of the cover shown in FIG. 3.

(8) FIG. 8 is a perspective view of a cover in a first modified example.

(9) FIG. 9 is a perspective view of a cover in a second modified example.

(10) FIG. 10 is a plan view of the cover shown in FIG. 9.

DETAILED DESCRIPTION

Description of Embodiments of Present Disclosure

(11) The embodiments of the present disclosure will now be described.

(12) A connector according to the present disclosure, including [1] a connector housing including an opening, a cover closing the opening, and terminals held by the connector housing. Each of the terminals includes a connection part configured to be electrically connected to one of a plurality of mate terminals, the mate terminals differing from one another. The connection parts are arranged in a line in the connector housing. The cover includes a first surface exposed outside the connector housing and a second surface located at a rear side of the first surface. The second surface includes an exposed surface exposed from the opening to an inner side of the connector housing. The cover further includes at least one first reinforcement rib projecting from the first surface and at least one second reinforcement rib projecting from the second surface. When a first direction refers to a direction in which the connection parts are arranged, a second direction refers to a direction intersecting the first direction and extending from a basal end to a distal end of the connection parts, and a third direction refers to a direction orthogonal to the first direction and the second direction, the first reinforcement rib extends in the first direction, and the second reinforcement rib extends in the second direction between adjacent ones of the connection parts as viewed in the third direction.

(13) With this structure, the cover includes the first reinforcement rib and the second reinforcement rib, respectively arranged on the two surfaces, namely, the first surface and the second surface located at the rear side of the first surface. The first reinforcement rib projects from the first surface and extends in the first direction. The second reinforcement rib projects from the second surface and extends in the second direction, which intersects the first direction. This increases the rigidity of the cover as compared to a structure in which only one of the first surface and the second surface includes a reinforcement rib or a structure in which the first reinforcement rib and the second reinforcement rib extend in the same direction. Thus, when the cover receives an external force, deformation of the cover is restricted. [2] Preferably, the cover includes a sealing portion projecting from the second surface to surround the exposed surface as viewed in the third direction, the sealing portion is fitted into the opening, and at least one of the second reinforcement rib is greater than the sealing portion in a length of projection from the second surface.

(14) With this structure, the sealing portion also increases the rigidity of the cover. The sealing portion, fitted into the opening, readily limits entrance of liquid such as water into the connector housing from the opening. [3] Preferably, the cover includes a resin molding component, and as viewed in any of the second direction and the third direction, the second reinforcement rib is disposed between each two of the connection parts located adjacent to each other in the first direction.

(15) This structure allows the second reinforcement rib to be elongated in the third direction as compared to a structure in which the second reinforcement rib is disposed between the connection parts located adjacent to each other in the first direction as viewed in the third direction but is not disposed between the connection parts located adjacent to each other in the first direction as viewed in the second direction. Thus, the rigidity of the cover is further increased. [4] Preferably, the second surface extends wider than the exposed surface in opposite directions along the first direction, the connector housing includes two first guides extending in the third direction from two ends of the connector housing located at opposite sides in the first direction, the cover includes two second guides projecting from two parts of the second surface located outward from the exposed surface in the first direction, and the two second guides are fitted to the two first guides using a recess and a projection to restrict misalignment of the connector housing with the cover in the first direction and the second direction.

(16) With this structure, when the cover is disposed on the connector housing, the first guides are fitted to the second guides using a recess and a projection. Thus, the cover is easily positioned on the connector housing in the first direction and the second direction. This facilitates the coupling of the cover to the connector housing. [5] Preferably, the cover includes two first reinforcement parts and/or two second reinforcement parts, the two first reinforcement parts extend on the second surface from the two second guides to a first end of the cover in the second direction, and the two second reinforcement parts extend on the second surface from the two second guides to a second end of the cover in the second direction.

(17) With this structure, the rigidity of the cover is increased by the second guides and the first reinforcement parts and/or the second reinforcement parts. [6] Preferably, the cover includes a peripheral wall projecting from the first surface and extending along a peripheral edge of the first surface.

(18) With this structure, the rigidity of the cover is further increased by the peripheral wall. [7] Preferably, the peripheral wall annularly extends along the peripheral edge of the first surface.

(19) With this structure, including the annular peripheral wall, the rigidity of the cover is more likely to be increased than with a structure in which the peripheral wall is not annular. [8] Preferably, the at least one second reinforcement rib includes second reinforcement ribs, and the cover includes at least one joint rib that joins ones of the second reinforcement ribs that are located adjacent to each other in the first direction.

(20) With this structure, the rigidity of the second reinforcement ribs is increased. Thus, the rigidity of the cover is further increased. [9] Preferably, the cover includes two of the first reinforcement rib.

(21) With this structure, the rigidity of the cover is increased as compared to a structure in which the cover includes only one first reinforcement rib. In addition, the shape of the cover is simplified as compared to a structure in which the cover includes three or more first reinforcement ribs. Furthermore, the material forming the cover is reduced as compared to a structure in which the cover includes three or more first reinforcement ribs. [10] Preferably, the cover includes two end surfaces located in the second direction, at least one of the two end surfaces includes an outward surface extending in the first direction, and the cover includes a sealing portion projecting from the second surface to surround the exposed surface as viewed in the third direction, the sealing portion is fitted into the opening, an annular sealant sealing a gap between the connector housing and an outer surface of the sealing portion that faces the connector housing, and a protrusion protruding from at least one of the outward surface in the second direction or a direction opposite to the second direction.

(22) With this structure, in addition to the sealing portion, the sealant limits entrance of liquid into the connector housing from the opening of the connector housing. In addition, when the cover is not coupled to the connector housing and is placed on a surface of a ground, a floor, or the like, the cover may be located so that the outward surface, from which the protrusion protrudes, faces the surface. In this case, the cover is located on the surface of a ground, a floor, or the like, so that the protrusion, protruding from the outward surface, spaces the outward surface apart from the surface. That is, the cover is partially lifted by the protrusion and is located on the surface. Thus, the sealing portion is likely to be disposed at a position separated upward from the surface of a ground, a floor, or the like. This limits contact of the sealant, which is disposed on the outer surface of the sealing portion, with the surface of a ground, a floor, or the like. Accordingly, objects such as sand or dust on the surface are less likely to collect on the sealant.

Detailed Description of Embodiments of Present Disclosure

(23) Specific examples of a connector according to the present disclosure will be described below with reference to the drawings. The present disclosure is not limited to those exemplified and is shown by the scope of the claims. It is intended to include all modifications within the meaning and range equivalent to the scope of the claims. The dimensional proportion of a component may not be

drawn to scale or may differ between drawings.

(24) An embodiment of a connector will now be described.

(25) FIG. 1 shows the present embodiment of a connector **10** configured to be electrically connected to an electrical device, which is mounted on a vehicle and is not shown in the drawings. An example of the electrical device is a wheel driving motor used as a power source for the vehicle to travel. However, the electrical device is not limited to the motor. The electrical device is accommodated in a case, which is not shown in the drawings. The case has a mount hole that connects the inside and the outside of the case. A terminal mount includes mate terminals **9** and is disposed in the mount hole. The mate terminals **9** are shown in only FIG. 7. The terminal mount includes, for example, three mate terminals **9**. The mate terminals **9** are each electrically connected to the electrical device inside the case. When the connector **10** is disconnected from the terminal mount, the mate terminals **9** are exposed to the outside of the case from the terminal mount.

(26) As shown in FIGS. 1 and 2, the connector **10** includes a connector housing **12** including an opening **11**, a cover **13** closing the opening **11**, and terminals **14** held by the connector housing **12**. In the description hereafter, the connector housing **12** is referred to as the housing **12**.

(27) Terminal **14**

(28) As shown in FIGS. 2 and 7, in the present embodiment, the connector **10** includes three terminals **14**, which are equal in number to the mate terminals **9**. The number of the terminals **14** in the connector **10** is not limited to three and may be changed to any plural number. The terminals **14** are formed from a metal material. The material of the terminals **14** may be, for example, copper, a copper alloy, aluminum, an aluminum alloy, or other metal materials. Each of the terminals **14** is electrically connected to a different one of the mate terminals **9**.

(29) Each terminal **14** includes a connection part **21**. Each connection part **21** is electrically connected to a different one of the mate terminals **9**. The connection part **21** is a part of the terminal **14** arranged on the mate terminal **9**. The terminal **14** includes the connection part **21** on an end of the terminal **14** projecting to an inner side of the housing **12**. Each terminal **14** includes a terminal main body **22** held by the housing **12**. In the terminal **14**, the connection part **21** extends from an end of the terminal main body **22** that projects to the inner side of the housing **12**. In the terminal **14**, the connection part **21** includes two ends in a direction in which the connection part **21** projects from the terminal main body **22**, and one of the two ends located farther from the terminal main body **22** defines a distal end of the connection part **21**. Also, in the terminal **14**, one of the two ends of the connection part **21**, in a direction in which the connection part **21** projects from the terminal main body **22**, located closer to the terminal main body **22** defines a basal end of the connection part **21**.

(30) In the present embodiment, the connection parts **21** of the terminals **14** are identical in shape to each other. In an example, the connection part **21** has the form of a plate including a through hole. The connection part **21** includes a connection surface **23** that contacts a corresponding one of the mate terminals **9**. In an example, the connection surface **23** is a side surface of the connection part **21** in a thickness-wise direction of the connection part **21**.

(31) The connection parts **21** are arranged in a line in the housing **12**. In an example, the connection parts **21** are arranged in a line so that the connection surfaces **23** of the connection parts **21** are coplanar with each other. The connection parts **21** are arranged so that the directions extending from the basal ends to the distal ends of the connection parts **21** are parallel to each other. Adjacent ones of the connection parts **21** are separated in a direction in which the connection parts **21** are arranged.

(32) The arrangement direction of the connection parts **21** refers to a first direction X. The first direction X may refer to the direction indicated by the arrow of the x-axis in the drawings. When the three connection parts **21** are referred to as the connection parts **21a** to **21c**, the connection parts **21a** to **21c** are arranged in the order of the connection part **21a**, the connection part **21b**, and the connection part **21c** in the first direction X. The direction extending from the basal end toward the

distal end of the connection part **21** is referred to as a second direction Y. The second direction Y is a direction intersecting the first direction X. More specifically, when the directions extending from the basal ends to the distal ends of the connection parts **21** are parallel to each other, the direction extending from the basal end toward the distal end of each connection part **21** as viewed in the first direction X is referred to as the second direction Y. A direction that is orthogonal to the first direction X and the second direction Y is referred to as a third direction Z. In an example, as viewed in the first direction X, when the second direction Y extends from the left toward the right, the third direction Z extends from the lower side to the upper side. In the description hereafter, the first direction X, the second direction Y, and the third direction Z refer to the first direction X, the second direction Y, and the third direction Z in the connector **10** in an assembled state.

(33) Electric wires **25** are separately and electrically connected to the terminals **14**. Each electric wire **25** is electrically connected to a part of the terminal **14** that is separated from the connection part **21**. In an example, the electric wire **25** is a coated wire that includes a core wire, which is formed of a conductor and an insulating cover, which covers the circumference of the core wire. The electric wire **25** includes an end connected to the terminal **14** and the opposite end electrically connected to an electrical device differing from the electrical device that is electrically connected to the connector **10** by the terminal mount. The opposite end of the electric wire **25** from the end connected to the terminal **14** is electrically connected to, for example, an inverter that generates alternating-current power from direct current power of a battery. Alternatively, the end of the electric wire **25** may be electrically connected to an electrical device differing from an inverter.

(34) Connector Housing **12**

(35) In an example, the housing **12** is a resin mold component. In an example, the terminals **14** are partially disposed in the resin forming the housing **12** through insert molding so that the housing **12** holds the terminals **14**. Alternatively, the terminals **14** may be formed separately from the housing **12** and then fixed to the housing **12**.

(36) The housing **12** includes a housing main body **31** in which the connection parts **21a** to **21c** are disposed. The housing main body **31** is tubular. In an example, as viewed in an axial direction of the housing main body **31**, the housing main body **31** has a low-profile shape such that the housing main body **31** is elongated in the first direction X and is short in the second direction Y. The ends of the terminals **14** including the connection parts **21a** to **21c** project from an inner circumferential surface of the housing main body **31** to the cavity located at an inner side of the housing main body **31**. In the present embodiment, the thickness-wise direction of the connection parts **21a** to **21c** is parallel to the axial direction of the housing main body **31**.

(37) The housing main body **31** includes the opening **11**. The housing main body **31** includes the opening **11** located in an end of the housing main body **31** in the third direction Z. In an example, the opening **11** corresponds to one of two axial end openings in the housing main body **31**. In the housing main body **31**, the opening opposite to the opening **11** is an insertion opening **32** that is disposed in the terminal mount or the case.

(38) As shown in FIGS. **1** and **2**, the housing **12** may include an electric wire holder **33** that holds the ends of the electric wires **25** electrically connected to the terminals **14**. In an example, the electric wire holder **33** is formed integrally with the housing main body **31**. The electric wire holder **33** projects from the housing main body **31** to an outer side of the housing main body **31**. Portions of the electric wires **25** and the terminals **14** electrically connected to each other are held inside the electric wire holder **33**.

(39) The housing **12** may include first guides **34a** and **34b** extending in the third direction Z from two ends of the housing **12** located at opposite sides in the first direction X.

(40) The housing main body **31** includes a first end in the first direction X. The first guide **34a** projects from the first end in the first direction X and extends in the third direction Z. The first guide **34a** includes a guide recess **35a**. The guide recess **35a** is recessed in a direction opposite to the first direction X from a distal surface of the first guide **34a** in the first direction X. The guide

recess **35a** extends through the first guide **34a** in the third direction Z. In an example, the guide recess **35a** has a quadrangular cross section that is cut along a plane orthogonal to the third direction Z. The cross-sectional shape of the guide recess **35a**, which is cut along the plane orthogonal to the third direction Z, is not limited to a quadrangle and may be any shape, such as a semicircle. The cross sectional shape of the first guide **34a** that is cut along the plane orthogonal to the third direction Z is uniform in the third direction Z.

(41) The housing main body **31** includes a second end in the first direction X. The first guide **34b** projects from the second end in the opposite direction of the first guide **34a** and extends in the third direction Z. The first guide **34b** includes a guide recess **35b**. The guide recess **35b** is recessed in the first direction X from a distal surface of the first guide **34b**, which is located in the opposite direction of the first direction X. The guide recess **35b** extends through the first guide **34b** in the third direction Z. In an example, the guide recess **35b** has a quadrangular cross section that is cut along a plane orthogonal to the third direction Z. The cross sectional shape of the guide recess **35b**, which is cut along the plane orthogonal to the third direction Z, is not limited to a quadrangle and may be any shape such as a semicircle. The cross sectional shape of the first guide **34b** that is cut along the plane orthogonal to the third direction Z is uniform in the third direction Z.

(42) Cover **13**

(43) In an example, the cover **13** is a resin mold component. In an example, the cover **13** is formed from a resin material having a lower conductivity than the terminals **14**. The cover **13** includes a closing portion **41** that closes the opening **11**. The closing portion **41** has the form of a plate that is larger than the opening **11** as viewed in the third direction Z. In an example, the closing portion **41** has the form of a plate such that the dimension in the first direction X is larger than the dimension in the second direction Y (width). The cover **13** is disposed on the housing **12** so that the opening **11** is covered by the closing portion **41**. When the cover **13** is disposed on the housing **12**, the thickness-wise direction of the closing portion **41** conforms to the third direction Z. In the cover **13**, the thickness-wise direction of the closing portion **41** also conforms to the axial direction of the housing main body **31**. However, the thickness-wise direction of the closing portion **41** does not necessarily have to exactly conform to the third direction Z. Also, the thickness-wise direction of the closing portion **41** does not necessarily have to exactly conform to the axial direction of the housing main body **31**.

(44) As shown in FIG. 7, the cover **13** includes a first surface **42** exposed outside the housing **12** and a second surface **43** located at a rear side of the first surface **42**. FIG. 7 is a cross-sectional view taking along line 7-7 in FIG. 3 showing a cross section of the cover **13** that is cut along a plane orthogonal to the second direction Y.

(45) In an example, the first surface **42** is a side surface of the closing portion **41** located at a side opposite to the opening **11**. The first surface **42** is planar and orthogonal to the third direction Z. The first surface **42** does not necessarily have to be exactly orthogonal to the third direction Z.

(46) In an example, the second surface **43** is a side surface of the closing portion **41** located at the rear of the first surface **42**. The second surface **43** faces the opening **11**. The second surface **43** includes an exposed surface **43a** exposed from the opening **11** to an inner side of the housing **12**. The second surface **43** may extend from the exposed surface **43a** in opposite directions along the first direction X. In an example, the second surface **43** includes the exposed surface **43a** and a non-facing surface **43b** that is located around the exposed surface **43a** and does not face the opening **11**. The second surface **43** is planar and orthogonal to the third direction Z. The second surface **43** does not necessarily have to be exactly orthogonal to the third direction Z. In an example, the exposed surface **43a** and the non-facing surface **43b** are coplanar with each other. However, the exposed surface **43a** and the non-facing surface **43b** may be located at different positions in the third direction Z. The second surface **43** may include only the exposed surface **43a**.

(47) As shown in FIGS. 4 to 7, the cover **13** may include a sealing portion **44** projecting from the second surface **43** to surround the exposed surface **43a** as viewed in the third direction Z. The

sealing portion **44** may be fitted into the opening **11**. The sealing portion **44** is tubular and projects from the second surface **43** and extends in a direction opposite to the third direction Z. The sealing portion **44** is slightly smaller than the inner circumferential surface of the housing main body **31** as viewed in the third direction Z so that the sealing portion **44** is fitted into the housing main body **31** from the opening **11**. Thus, in the present embodiment, as viewed in the third direction Z, the sealing portion **44** has a low-profile shape such that the sealing portion **44** is elongated in the first direction X and is short in the second direction Y. As viewed in the third direction Z, the exposed surface **43a** is a part of the second surface **43** located inward from the sealing portion **44**.

(48) The sealing portion **44** includes side walls **44a** and **44b** facing each other in the second direction Y. The side wall **44a** defines a first end of the sealing portion **44** in the second direction Y. The side wall **44b** defines a second end of the sealing portion **44** in the second direction Y. The side walls **44a** and **44b** extend in the first direction X as viewed in the third direction Z.

(49) The sealing portion **44** may include a sealing groove **44c** extending in an outer circumferential surface of the sealing portion **44**. In an example, the sealing groove **44c** extends annularly as viewed in the third direction Z. The sealing groove **44c** is open toward the outer circumference of the sealing portion **44**.

(50) The cover **13** may include an annular sealant **44d** sealing a gap between the housing **12** and the outer surface of the sealing portion **44** that faces the housing **12**. In an example, the sealant **44d** seals the gap between the outer surface of the sealing portion **44** and the inner surface of the housing main body **31** opposed to the outer surface of the sealing portion **44**. In an example, the sealant **44d** is disposed in the sealing groove **44c**. In an example, a rubber ring may be used as the sealant **44d**. Besides a rubber ring, the sealant **44d** may be any annular sealant that extends around the outer circumference of the sealing portion **44**. When the sealing portion **44** is fitted into the housing main body **31**, the sealant **44d** is in hermetical contact with the inner surface of the housing main body **31** and an inner wall surface of the sealing groove **44c**. When liquid such as water enters the inside of the housing main body **31** from the opening **11**, the sealant **44d** limits movement of the liquid toward the terminal mount to which the connector **10** is connected.

(51) As shown in FIGS. **1** to **4** and **7**, the cover **13** includes at least one first reinforcement rib **45** projecting from the first surface **42**. In the present embodiment, the cover **13** includes two first reinforcement ribs **45**. The number of first reinforcement ribs **45** is not limited to two, and the cover **13** may include one or three or more first reinforcement ribs **45**.

(52) Each first reinforcement rib **45** projects from the first surface **42** in the third direction Z, that is, in a direction away from the opening **11**. The first reinforcement rib **45** also extends in the first direction X as viewed in the third direction Z. In an example, the first reinforcement rib **45** extends from a first end to a second end of the first surface **42** in the first direction X. In other words, the first reinforcement rib **45** extends from a first end to a second end of the closing portion **41** in the first direction X. The first reinforcement ribs **45** are integrated with the closing portion **41**. The two first reinforcement ribs **45** are separated in the second direction Y. For example, the two first reinforcement ribs **45** extend parallel to each other. The first reinforcement ribs **45** do not necessarily have to be parallel to each other as long as the first reinforcement ribs **45** extend in the first direction X as viewed in the third direction Z.

(53) In an example, the first reinforcement rib **45** has a quadrangular cross section that is cut along a plane orthogonal to the first direction X. The shape of a cross section of the first reinforcement rib **45** cut along a plane orthogonal to the first direction X is not limited to a quadrangle and may be any shape projecting from the first surface **42**. In an example, the cross-sectional shape of the first reinforcement rib **45** cut along a plane orthogonal to the first direction X may be a polygon including a quadrangle, a semicircle, or a partial cutaway circle.

(54) The cover **13** may include a peripheral wall **46** projecting from the first surface **42** and extending along the peripheral edge of the first surface **42**. In an example, the peripheral wall **46** annularly extends along the peripheral edge of the first surface **42**. The term “annular” as used in

this description may refer to any structure that forms a loop or a continuous shape with no ends. “Annular” shapes include but are not limited to a circular shape, an elliptic shape, and a polygonal shape with sharp or rounded corners.

(55) The peripheral wall **46** projects from the first surface **42** in the third direction **Z**. In an example, the peripheral wall **46** has the form of a frame such that the dimension in the first direction **X** is larger than the dimension in the second direction **Y** as viewed in the third direction **Z**. The peripheral wall **46** is integrated with the closing portion **41**. In an example, the peripheral wall **46** has a quadrangular cross section that is cut along a plane orthogonal to the direction in which the peripheral wall **46** extends. The shape of the cross section of the peripheral wall **46** is not limited to a quadrangle and may be any shape projecting from the first surface **42**. In an example, the cross-sectional shape of the peripheral wall **46** may be a polygon including a quadrangle, a semicircle, or a partial cutaway circle.

(56) In an example, as viewed in the third direction **Z**, the first reinforcement ribs **45** are located inward from the peripheral wall **46**. Each first reinforcement rib **45** has opposite ends in the first direction **X** that are joined to the peripheral wall **46**. More specifically, the first reinforcement rib **45** is integrated with the peripheral wall **46**. In an example, the first reinforcement ribs **45** and the peripheral wall **46** project from the first surface **42** by the same length. The first reinforcement ribs **45** and the peripheral wall **46** may project from the first surface **42** by different lengths.

(57) As shown in FIGS. **4** to **7**, the cover **13** includes at least one second reinforcement rib **47** projecting from the second surface **43**. In an example, the number of the second reinforcement ribs **47** included in the cover **13** is less than the number of the connection parts **21** by one. More specifically, in the present embodiment, since the connector **10** includes three terminals **14**, the cover **13** includes two second reinforcement ribs **47**. The number of the second reinforcement ribs **47** included in the cover **13** is not limited to two and may be any number that is greater than or equal to one. The number of the second reinforcement ribs **47** does not have to be less than the number of terminals **14** by one. In an example, when the connector **10** includes four or more terminals **14**, the number of the second reinforcement ribs **47** may be less than the number of the terminals **14** by two.

(58) In the present embodiment, the cover **13** is formed from a resin material having a lower conductivity than the terminals **14**. Thus, the second reinforcement ribs **47** are less conductive than the terminals **14**.

(59) Each second reinforcement rib **47** projects from the second surface **43** in a direction opposite to the third direction **Z**, that is, in a direction away from the first surface **42**. Also, as viewed in the third direction **Z**, the second reinforcement rib **47** extends in the second direction **Y** between adjacent ones of the connection parts **21**. In an example, as viewed in the third direction **Z**, one second reinforcement rib **47** is disposed between each two of the connection parts **21** located adjacent to each other in the first direction **X**. In other words, as viewed in the third direction **Z**, the connection parts **21** are disposed at opposite sides of each second reinforcement rib **47** in the first direction **X**. More specifically, as viewed in the third direction **Z**, one of the second reinforcement ribs **47** is disposed between the connection part **21a** and the connection part **21b**, which are located adjacent to each other in the first direction **X**. In the same manner, as viewed in the third direction **Z**, one of the second reinforcement ribs **47** is disposed between the connection part **21b** and the connection part **21c**, which are located adjacent to each other in the first direction **X**.

(60) In an example, each second reinforcement rib **47** projects from the exposed surface **43a** in a direction opposite to the third direction **Z**. The second reinforcement ribs **47** are integrated with the closing portion **41**. As viewed in the third direction **Z**, each second reinforcement rib **47** is flat and extends in the second direction **Y**. In an example, the second reinforcement rib **47** is flat and is orthogonal to the first direction **X**. However, the second reinforcement rib **47** does not necessarily have to be orthogonal to the first direction **X**. The two second reinforcement ribs **47** are separated in the first direction **X**. In an example, the two second reinforcement ribs **47** extend parallel to each

other. The two second reinforcement ribs **47** do not necessarily have to be parallel to each other as long as the second reinforcement ribs **47** extend in the second direction Y as viewed in the third direction Z.

(61) In an example, as viewed in the third direction Z, the second reinforcement ribs **47** are located inward from the outer circumferential surface of the sealing portion **44**. In an example, each second reinforcement rib **47** has opposite ends in the second direction Y that are joined to the sealing portion **44**. More specifically, the second reinforcement rib **47** includes a first end in the second direction Y joined to the side wall **44a**. The second reinforcement rib **47** includes a second end in the second direction Y joined to the side wall **44b**. Thus, the second reinforcement ribs **47** may be integrated with the sealing portion **44**.

(62) As shown in FIG. 7, each second reinforcement rib **47** projects from the second surface **43** beyond the connection parts **21**. More specifically, distance D1 between the second surface **43** and the distal end of the second reinforcement rib **47** in the third direction Z is greater than distance D2 between the second surface **43** and each connection surface **23** in the third direction Z. Thus, the second reinforcement rib **47** is disposed between each ones of the connection parts **21** located adjacent to each other in the first direction X. In an example, in the first direction X, one of the second reinforcement ribs **47** is disposed between the connection part **21a** and the connection part **21b**, which are located adjacent to each other. In an example, when the connector **10** is viewed in any of the second direction Y and the third direction Z, the second reinforcement ribs **47** are disposed between the connection parts **21** located adjacent to each other in the first direction X.

(63) The at least one second reinforcement rib **47** may be greater than the sealing portion **44** in a length of projection from the second surface **43**. In an example, each second reinforcement rib **47** is greater than the sealing portion **44** in a length of projection from the second surface **43**. Thus, the second reinforcement rib **47** is greater than the sealing portion **44** in a length of projection from the second surface **43** in a direction opposite to the third direction Z. Therefore, distance D1 is greater than distance D3 that is between the second surface **43** and the distal end of the sealing portion **44** in the third direction Z. The two second reinforcement ribs **47** may be equal in the length of projection from the second surface **43**. The second reinforcement ribs **47** may differ in the length of projection from the second surface **43**.

(64) As shown in FIGS. 4 to 7, the cover **13** may include at least one joint rib **48** that joins ones of the second reinforcement ribs **47** that are located adjacent to each other in the first direction X. The “second reinforcement ribs **47** that are located adjacent to each other in the first direction X” refers to second reinforcement ribs **47** located next to each other in the first direction X when only the second reinforcement ribs **47** are focused on. In an example, the cover **13** includes one joint rib **48**. The joint rib **48** joins the two second reinforcement ribs **47** located at opposite sides of the connection part **21b** as viewed in the third direction Z. More specifically, as viewed in the third direction Z, the joint rib **48** joins the second reinforcement rib **47** that is disposed between the connection part **21a** and the connection part **21b** and the second reinforcement rib **47** that is disposed between the connection part **21b** and the connection part **21c**.

(65) The joint rib **48** is located closer to the connection parts **21** than the second surface **43** in the third direction Z. As viewed in the third direction Z, the joint rib **48** extends in the first direction X. In an example, the joint rib **48** is disposed on a distal surface of the side wall **44a**. The joint rib **48** projects from the distal surface of the side wall **44a** in a direction opposite to the third direction Z. The joint rib **48** joints the first ends, in the second direction Y, of the two second reinforcement ribs **47** located adjacent to each other in the first direction X. The joint rib **48** is integrated with the sealing portion **44** and the two second reinforcement ribs **47** joined by the joint rib **48**. The distal end of the joint rib **48** and the distal end of the second reinforcement rib **47** may be located at the same position in the third direction Z. The distal end of the joint rib **48** and the distal end of the second reinforcement rib **47** may be located at different positions in the third direction Z.

(66) The cover **13** may include at least one auxiliary rib **49** integrated with the second

reinforcement ribs **47**. The auxiliary rib **49** is integrated with the second end of the second reinforcement rib **47** in the second direction Y. In an example, the auxiliary rib **49** may be disposed on each of the second reinforcement ribs **47**. Each auxiliary rib **49** extends in the first direction X as viewed in the third direction Z. Each auxiliary rib **49** and the second reinforcement rib **47** that is integrated with the auxiliary rib **49** are T-shaped as viewed in the third direction Z. In an example, the auxiliary rib **49** is disposed on a distal surface of the side wall **44b**. The auxiliary rib **49** projects from the distal surface of the side wall **44b** in a direction opposite to the third direction Z. The auxiliary rib **49** is integrated with the sealing portion **44**. The distal end of the auxiliary rib **49** and the distal end of the second reinforcement ribs **47** may be located at the same position in the third direction Z. The distal end of the auxiliary rib **49** and the distal end of the second reinforcement rib **47** may be located at different positions in the third direction Z.

(67) As shown in FIGS. **4** to **6**, the cover **13** may include a projection wall **51a** in a first end region of the cover **13** in the first direction X. The cover **13** may also include a projection wall **51b** in a second end region of the cover **13** in the first direction X. The projection wall **51a** projects from a portion of the second surface **43** separated from the exposed surface **43a** in the first direction X. In an example, the projection wall **51a** projects from the first end, in the first direction X, of the second surface **43** in a direction opposite to the third direction Z. The projection wall **51b** projects from a portion of the second surface **43** separated from the exposed surface **43a** in a direction opposite to the first direction X. In an example, the projection wall **51b** projects from the second end, in the first direction X, of the second surface **43** in a direction opposite to the third direction Z. The projection walls **51a** and **51b** are integrated with the closing portion **41**. In an example, the projection walls **51a** and **51b** are flat and orthogonal to the first direction X.

(68) The cover **13** may include a fixing portion **52a** projecting from the projection wall **51a**. The cover **13** may also include a fixing portion **52b** projecting from the projection wall **51b**. The fixing portion **52a** projects from a distal end of the projection wall **51a** in the first direction X. The fixing portion **52a** includes a fixing hole **53a** extending through the fixing portion **52a** in the third direction Z. The fixing portion **52b** projects from a distal end of the projection wall **51b** in a direction opposite to the first direction X. The fixing portion **52b** includes a fixing hole **53b** extending through the fixing portion **52b** in the third direction Z.

(69) The cover **13** may include rib walls **56a** and **56b** projecting from the second surface **43** and integrated with the projection wall **51a**. Also, the cover **13** may include rib walls **56c** and **56d** projecting from the second surface **43** and integrated with the projection wall **51b**.

(70) As viewed in the third direction Z, the rib walls **56a** and **56b** are located at a position separated from the exposed surface **43a** in the first direction X. The rib wall **56a** projects from the first end, in the second direction Y, of the second surface **43** in a direction opposite to the third direction Z. The rib wall **56b** projects from the second end, in the second direction Y, of the second surface **43** in a direction opposite to the third direction Z. The rib wall **56a** includes a first end in the first direction X that is joined to the projection wall **51a**. The rib wall **56b** includes a first end in the first direction X that is joined to the projection wall **51a**. In an example, the rib walls **56a** and **56b** are flat and orthogonal to the second direction Y. Each of the rib walls **56a** and **56b** projects from the second surface **43** by a length that decreases as the projection wall **51a** becomes farther in a direction opposite to the first direction X.

(71) As viewed in the third direction Z, the rib walls **56c** and **56d** are located at a position separated from the exposed surface **43a** in a direction opposite to the first direction X. The rib wall **56c** projects from the first end, in the second direction Y, of the second surface **43** in a direction opposite to the third direction Z. The rib wall **56d** projects from the second end, in the second direction Y, of the second surface **43** in a direction opposite to the third direction Z. The rib wall **56c** includes a second end in the first direction X that is joined to the projection wall **51b**. The rib wall **56d** includes a second end in the first direction X that is joined to the projection wall **51b**. In an example, the rib walls **56c** and **56d** are flat and orthogonal to the second direction Y. Each of the

rib walls **56c** and **56d** projects from the second surface **43** by a length that decreases as the projection wall **51b** becomes farther in the first direction X.

(72) As shown in FIGS. **2** and **4** to **6**, the cover **13** may include second guides **57a** and **57b** respectively projecting from a portion of the second surface **43** separated from the exposed surface **43a** in the first direction X and a portion of the second surface **43** separated from the exposed surface **43a** in a direction opposite to the first direction X. The second guide **57a** projects in a direction opposite to the third direction Z from a portion of the non-facing surface **43b** separated from the exposed surface **43a** in the first direction X. The second guide **57a** is located at a position where the second guide **57a** is fitted into the guide recess **35a** in a direction opposite to the third direction Z when the cover **13** is disposed on the housing **12**. The second guide **57b** projects in a direction opposite to the third direction Z from a portion of the non-facing surface **43b** separated from the exposed surface **43a** in a direction opposite to the first direction X. The second guide **57b** is located at a position where the second guide **57b** is fitted into the guide recess **35b** in a direction opposite to the third direction Z when the cover **13** is disposed on the housing **12**.

(73) The shape of the second guide **57a** is in conformance with the shape of wall surfaces of the guide recess **35a**. In an example, the second guide **57a** and the guide recess have a quadrangular cross section that is cut along a plane orthogonal to the third direction Z and is similar to each other. The shape of a cross section of the second guide **57a** cut along a plane orthogonal to the third direction Z is not limited to a quadrangle and may be changed in accordance with the shape of a cross section of the guide recess **35a** cut along a plane orthogonal to the third direction Z.

(74) The shape of the second guide **57b** is in conformance with the shape of wall surfaces of the guide recess **35b**. In an example, the second guide **57b** and the guide recess have a quadrangular cross section that is cut along a plane orthogonal to the third direction Z and is similar to each other. The shape of a cross section of the second guide **57b** cut along a plane orthogonal to the third direction Z is not limited to a quadrangle and may be changed in accordance with the shape of a cross section of the guide recess **35b** cut along a plane orthogonal to the third direction Z.

(75) In an example, as viewed in the third direction Z, the second guides **57a** and **57b** extend in the second direction Y. The second guides **57a** and **57b** may be flat and orthogonal to the first direction X. The second guides **57a** and **57b** do not have to be exactly orthogonal to the first direction X.

(76) When the cover **13** is disposed on the housing **12**, the second guide **57a** is fitted into the guide recess **35a** in a direction opposite to the third direction Z. Thus, the second guide **57a** and the first guide **34a** are fitted using a recess and a projection. When the cover **13** is disposed on the housing **12**, the second guide **57b** is fitted into the guide recess **35b** in a direction opposite to the third direction Z. Thus, the second guide **57b** and the first guide **34b** are fitted using a recess and a projection. The recess-projection fitting of the second guides **57a** and **57b** to the first guides **34a** and **34b** restricts misalignment of the housing **12** with the cover **13** in the first direction X and the second direction Y.

(77) As shown in FIGS. **4** to **6**, the cover **13** may include first reinforcement parts **58a** and **58b** extending on the second surface **43** from the second guides **57a** and **57b** to a first end of the cover **13** in the second direction Y. The cover **13** may also include second reinforcement parts **59a** and **59b** extending on the second surface **43** from the second guides **57a** and **57b** to a second end of the cover **13** in the second direction Y.

(78) The first reinforcement part **58a** and the second reinforcement part **59a** project from a portion of the second surface **43** separated from the exposed surface **43a** in the first direction X. More specifically, the first reinforcement part **58a** and the second reinforcement part **59a** project in a direction opposite to the third direction Z from a portion of the non-facing surface **43b** separated from the exposed surface **43a** in the first direction X. As viewed in the third direction Z, the first reinforcement part **58a** and the second reinforcement part **59a** extend in the second direction Y. In an example, the first reinforcement part **58a** and the second reinforcement part **59a** are flat and are orthogonal to the first direction X. The first reinforcement part **58a** and the second reinforcement

part **59a** do not have to be exactly orthogonal to the first direction X. The first reinforcement part **58a** includes a second end in the second direction Y joined to the second guide **57a**. The first reinforcement part **58a** includes a first end in the second direction Y joined to the rib wall **56a**. The second reinforcement part **59a** includes a first end in the second direction Y joined to the second guide **57a**. The second reinforcement part **59a** includes a second end in the second direction Y joined to the rib wall **56b**.

(79) The first reinforcement part **58b** and the second reinforcement part **59b** project from a portion of the second surface **43** separated from the exposed surface **43a** in a direction opposite to the first direction X. More specifically, the first reinforcement part **58b** and the second reinforcement part **59b** project in a direction opposite to the third direction Z from a portion of the non-facing surface **43b** separated from the exposed surface **43a** in a direction opposite to the first direction X. As viewed in the third direction Z, the first reinforcement part **58b** and the second reinforcement part **59b** extend in the second direction Y. In an example, the first reinforcement part **58b** and the second reinforcement part **59b** are flat and are orthogonal to the first direction X. The first reinforcement part **58b** and the second reinforcement part **59b** do not have to be exactly orthogonal to the first direction X. The first reinforcement part **58b** includes a second end in the second direction Y joined to the second guide **57b**. The first reinforcement part **58b** includes a first end in the second direction Y joined to the rib wall **56c**. The second reinforcement part **59b** includes a first end in the second direction Y joined to the second guide **57b**. The second reinforcement part **59b** includes a second end in the second direction Y joined to the rib wall **56d**.

(80) The cover **13** may include third reinforcement ribs **61a** and **61b** projecting from the second surface **43**. The third reinforcement rib **61a** extends from the second guide **57a** to the projection wall **51a**. The third reinforcement rib **61a** includes a second end in the first direction X joined to the second guide **57a**. The third reinforcement rib **61a** includes a first end in the first direction X joined to the projection wall **51a**. The third reinforcement rib **61b** extends from the second guide **57b** in the projection wall **51b**. The third reinforcement rib **61b** includes a first end in the first direction X joined to the second guide **57b**. The third reinforcement rib **61b** includes a second end in the first direction X joined to the projection wall **51b**. As viewed in the third direction Z, the third reinforcement ribs **61a** and **61b** extend in the first direction X. In an example, the third reinforcement ribs **61a** and **61b** are flat and orthogonal to the second direction Y. The third reinforcement ribs **61a** and **61b** do not have to be exactly orthogonal to the second direction Y.

(81) Connection of Connector **10** to Terminal Mount

(82) As shown in FIGS. **2** and **7**, when connecting the connector **10** to a terminal mount, a person first disposes the housing **12** on the terminal mount. At this time, the housing **12** is not disposed on the cover **13**, and the opening **11** is not closed by the cover **13**. Subsequently, a tool is inserted into the housing **12** from the opening **11** to electrically connect the terminals **14** to the mate terminals **9**. In an example, the terminal **14** is electrically connected to the mate terminal **9** by a bolt, inserted into the terminal **14** and the mate terminal **9**, and a nut, fastened to the bolt.

(83) Then, as shown in FIGS. **2**, **6**, and **7**, the cover **13** is disposed on the housing **12** to close the opening **11** with the cover **13**. When the cover **13** is disposed on the housing **12**, the sealing portion **44** is fitted into the housing main body **31** from the opening **11**. At this time, the second guide **57a** and the second guide **57b** are fitted into the guide recess **35a** of the first guide **34a** and the guide recess **35b** of the first guide **34b** in a direction opposite to the third direction Z. This positions the cover **13** on the housing **12** in the first direction X, the direction opposite to the first direction X, the second direction Y, and the direction opposite to the second direction Y. In addition, when the cover **13** is coupled to the housing **12**, inclination of the cover **13** from the housing **12** is limited. More specifically, when the cover **13** is coupled to the housing **12**, inclination of the axial direction of the housing main body **31** from the axial direction of the sealing portion **44** is limited.

(84) When the cover **13** is disposed on the housing **12**, the sealing portion **44** is fitted into the housing main body **31**. The second reinforcement ribs **47** are each disposed between the connection

parts **21** located adjacent to each other in the first direction X. Subsequently, the cover **13** is fixed to a case of the electrical device using the fixing hole **53a** and a bolt that is inserted into the fixing hole **53b** and is not shown in the drawings. This completes connection of the connector **10** to the terminal mount.

(85) The operation of the present embodiment will now be described.

(86) When the pressure of a case is increased by heat generated from the electrical device or the like, the pressure applied to the cover **13** from the inside of the case will be increased. In this case, the cover **13** receives an external force from the inside of the case acting to push the cover **13** outward from the case. More specifically, the cover **13** receives an external force acting to push the exposed surface **43a** outward from the case, that is, outward from the housing **12**. In this regard, in the present embodiment, the cover **13** includes the first reinforcement ribs **45**, projecting from the first surface **42**, and the second reinforcement ribs **47**, projecting from the second surface **43**. As viewed in the third direction Z, the first reinforcement ribs **45** and the second reinforcement ribs **47** extend in directions intersecting each other. This increases the rigidity of the cover **13**. Thus, deformation of the cover **13** is restricted.

(87) The advantages of the present embodiment will now be described. (1) The connector **10** includes the housing **12** including the opening **11**, the cover **13** closing the opening **11**, and the terminals **14** held by the housing **12**. Each terminal **14** includes a connection part **21**. Each connection part **21** is electrically connected to a different one of the mate terminals **9**. The connection parts **21** are arranged in a line in the housing **12**. The cover **13** includes the first surface **42**, exposed outside the housing **12**, and the second surface **43** located at the rear side of the first surface **42** and including the exposed surface **43a**, exposed from the opening **11** to the inner side of the housing **12**. The cover **13** further includes the two first reinforcement ribs **45** projecting from the first surface **42** and the two second reinforcement ribs **47** projecting from the second surface **43**. The first direction X refers to the direction in which the connection parts **21** are arranged. The second direction Y refers to a direction intersecting the first direction X and extending from the basal end to the distal end of each connection part **21**. The third direction Z refers to a direction orthogonal to the first direction X and the second direction Y. The first reinforcement ribs **45** extend in the first direction X. As viewed in the third direction Z, the second reinforcement ribs **47** each extend in the second direction Y between adjacent ones of the connection parts **21**.

(88) With this structure, the cover **13** includes the first reinforcement ribs **45** and the second reinforcement ribs **47**, respectively arranged on the two surfaces, namely, the first surface **42** and the second surface **43** located at the rear side of the first surface **42**. The first reinforcement ribs **45** project from the first surface **42** and extend in the first direction X. The second reinforcement ribs **47** project from the second surface **43** and extend in the second direction Y, which intersects the first direction X. This increases the rigidity of the cover **13** as compared to a structure in which only one of the first surface **42** and the second surface **43** includes reinforcement ribs or a structure in which the first reinforcement ribs and the second reinforcement ribs extend in the same direction. Thus, when the cover **13** receives an external force, deformation of the cover **13** is restricted.

(89) The restriction of deformation of the cover **13** limits formation of a gap between the cover **13** and the housing **12** around the opening **11**. As a result, even when the temperature becomes high in the inside of the case accommodating the electrical device to which the connector **10** is electrically connected by the terminal mount, entrance of liquid such as water into the housing **12** is limited. Ultimately, entrance of the liquid such as water into the case is limited. (2) The cover **13** includes the sealing portion **44** projecting from the second surface **43** to surround the exposed surface **43a** as viewed in the third direction Z and fitted into the opening **11**. Each second reinforcement rib **47** is greater than the sealing portion **44** in a length of projection from the second surface **43**.

(90) With this structure, the sealing portion **44** also increases the rigidity of the cover **13**. Thus, deformation of the cover **13** is further restricted. The sealing portion **44**, fitted into the opening **11**,

further decreases the gap between the sealing portion **44** and the inner circumferential edge of the opening **11**. Thus, liquid such as water is less likely to enter the housing **12** from the opening **11**. (91) In the present embodiment, the cover **13** includes the sealant **44d** sealing a gap between the housing **12** and the outer surface of the sealing portion **44** that faces the housing **12**. Thus, entrance of liquid such as water into the housing **12** from the opening **11** is limited by the sealant **44d**. (3) The cover **13** is a resin mold component. As viewed in any of the second direction Y and the third direction Z, the second reinforcement ribs **47** are disposed between the connection parts **21** located adjacent to each other in the first direction X.

(92) This structure allows the second reinforcement ribs **47** to be elongated in the third direction Z as compared to a structure in which the second reinforcement ribs **47** are each disposed between the connection parts **21** located adjacent to each other in the first direction X as viewed in the third direction Z but are not disposed between the connection parts **21** located adjacent to each other in the first direction X as viewed in the second direction Y. Thus, the rigidity of the cover **13** is further increased. Accordingly, deformation of the cover **13** is further restricted.

(93) The cover **13** is a resin mold component and thus may be formed from a resin material having a lower conductivity than the terminal **14**. In an example, when the terminal **14** is connected to the mate terminal **9**, at least one of the connection part **21** and the mate terminal **9** may be scraped and produce metal flakes. To inhibit formation of a short-circuit of the connection parts **21** located adjacent to each other in the first direction X because of the metal flakes, it is preferred that the connection parts **21** located adjacent to each other in the first direction X undergo an insulating process. When the cover **13** is a resin mold component formed from a resin material having a lower conductivity than the terminals **14**, the second reinforcement ribs **47**, which are part of the cover **13**, are also formed from the resin material having a lower conductivity than the terminals **14**. Thus, the conductivity of the second reinforcement ribs **47** is easily made lower than the conductivity of the terminals **14**. Electrical insulation of the connection parts **21** located adjacent to each other in the first direction X is obtained by the second reinforcement ribs **47** disposed between the connection parts **21**. This eliminates the need for a separate structure that inhibits the short-circuiting of the connection parts **21** located adjacent to each other in the first direction X. As a result, the number of components of the connector **10** is reduced. (4) The second surface **43** extends wider than the exposed surface **43a** in opposite directions in the first direction. The housing **12** includes the first guides **34a** and **34b** extending in the third direction Z from two ends of the housing **12** located at opposite sides in the first direction X. The cover **13** includes the second guides **57a** and **57b** projecting from two parts of the second surface **43** located at outer sides of the exposed surface **43a** in the first direction X. The recess-projection fitting of the second guides **57a** and **57b** to the first guides **34a** and **34b** restricts misalignment of the housing **12** with the cover **13** in the first direction X and the second direction Y.

(94) With this structure, when the cover **13** is disposed on the housing **12**, the first guides **34a** and **34b** are fitted to the second guides **57a** and **57b** using recesses and projections. Thus, the cover **13** is easily positioned on the housing **12** in the first direction X and the second direction Y. This facilitates the coupling of the cover **13** to the housing **12**. (5) The cover **13** includes the first reinforcement parts **58a** and **58b** extending on the second surface **43** from the second guides **57a** and **57b** to the first end of the cover **13** in the second direction Y. The cover **13** also includes the second reinforcement parts **59a** and **59b** extending on the second surface **43** from the second guides **57a** and **57b** to the second end of the cover **13** in the second direction Y.

(95) With this structure, the rigidity of the cover **13** is increased by the second guides **57a** and **57b**, the first reinforcement parts **58a** and **58b**, and the second reinforcement parts **59a** and **59b**. As a result, deformation of the cover **13** is further restricted. (6) The cover **13** includes the peripheral wall **46** projecting from the first surface **42** and extending along the peripheral edge of the first surface **42**. With this structure, the rigidity of the cover **13** is further increased by the peripheral wall **46**. As a result, deformation of the cover **13** is further restricted. (7) The peripheral wall **46**

annularly extends along the peripheral edge of the first surface **42**. With this structure, including the annular peripheral wall **46**, the rigidity of the cover **13** is more likely to be increased than with a structure in which the peripheral wall **46** is not annular. As a result, deformation of the cover **13** is further restricted. (8) The cover **13** includes a plurality of second reinforcement ribs **47**. The cover **13** further includes at least one joint rib **48** that joins the second reinforcement ribs **47** located adjacent to each other in the first direction X. With this structure, the rigidity of the second reinforcement ribs **47** is increased. Thus, the rigidity of the cover **13** is further increased. As a result, deformation of the cover **13** is further restricted. (9) The cover **13** includes the two first reinforcement ribs **45**. With this structure, the rigidity of the cover **13** is increased as compared to a structure in which the cover **13** includes only one first reinforcement rib **45**. In addition, the shape of the cover **13** is simplified as compared to a structure in which the cover **13** includes three or more first reinforcement ribs **45**. Furthermore, the material forming the cover **13** is reduced as compared to a structure in which the cover **13** includes three or more first reinforcement ribs **45**. This reduces the manufacturing costs of the cover **13**. (10) In the cover **13** of the present embodiment, the exposed surface **43a** does not include a reinforcement rib that extends in the first direction X. The exposed surface **43a** includes the second reinforcement ribs **47**, which are disposed between adjacent ones of the connection parts **21** in the first direction X as viewed in the third direction Z. Thus, even when the distance from the exposed surface **43a** to the connection parts **21** is decreased in the third direction Z, the second reinforcement ribs **47** do not contact the connection parts **21** in a direction opposite to the third direction Z. As described above, in the cover **13**, the second surface **43** does not include a reinforcement rib that has a possibility of contacting the connection parts **21** in a direction opposite to the third direction Z. This allows the second surface **43** to have a structure for increasing the rigidity of the cover **13** while limiting enlargement of the connector **10** in the third direction Z. (11) In the present embodiment, the cover **13** is formed from a resin material. Even when the cover **13** is formed from a resin material, the rigidity of the cover **13** is increased by the first reinforcement ribs **45** and the second reinforcement ribs. This eliminates the need for adding a separate metal part to the cover **13** or forming the cover **13** from metal to restrict deformation of the cover **13**. Thus, the manufacturing costs of the connector **10** are reduced. (12) Each joint rib **48** joints the first ends, in the second direction Y, of the two second reinforcement ribs **47** located adjacent to each other in the first direction X. As viewed in the third direction Z, the first end of the second reinforcement rib **47** in the second direction Y refers to the end located closer to the distal end of the connection part **21** than to the basal end. When the cover **13** is properly positioned on the housing **12**, the joint rib **48** is inserted between the inner circumferential surface of the housing **12** and the distal end of the connection part **21**. When the cover **13** is reversely disposed in the first direction X and improperly positioned on the housing **12**, the joint rib **48** contacts the terminal **14** in the vicinity of the basal end of the connection part **21** in a direction opposite to the third direction Z. The contact of the joint rib **48** with the terminal **14** hinders further movement of the cover **13** in a direction opposite to the third direction Z. Therefore, when the position of the cover **13** in the third direction Z differs from the proper position of the cover **13** with respect to the housing **12**, further movement of the cover **13** in a direction opposite to the third direction Z is hindered. This allows one to recognize that the cover **13** is improperly positioned on the housing **12**. As a result, erroneous coupling of the cover **13** to the housing **12** is avoided.

(96) The present embodiment may be modified as follows. The present embodiment and the following modified examples can be combined within a range where the combined modified examples remain technically consistent with each other.

(97) The number of joint ribs **48** is not limited to one. The cover **13** may include a plurality of joint ribs **48**. FIG. 8 shows a first modified example in which the cover **13** includes five second reinforcement ribs **47** arranged next to each other in the first direction X. In this case, the cover **13** may include two joint ribs **48**, each of which joins two of the second reinforcement ribs **47** located

at one of the opposite sides of the center second reinforcement rib **47**. In FIG. **8**, the same reference characters are given to those components that are the same as the corresponding components in the above embodiment.

(98) The joint ribs **48** may project from the distal surface of the side wall **44b**. When the cover **13** does not include the sealing portion **44**, the joint ribs **48** may project from the second surface **43** and join the second reinforcement ribs **47** located adjacent to each other in the first direction X. The joint rib **48** may be omitted from the cover **13**.

(99) The auxiliary rib **49** may be integrated with at least one second reinforcement rib **47**. In an example, of a plurality of second reinforcement ribs **47**, at least one second reinforcement rib **47** does not have to be integrated with the auxiliary rib **49**. The auxiliary rib **49** may be integrated with the first end of the second reinforcement rib **47** in the second direction Y and may be integrated with the second end of the second reinforcement rib **47** in the second direction Y. The auxiliary rib **49** may be omitted from the cover **13**.

(100) The peripheral wall **46** does not have to be annular. More specifically, the peripheral wall **46** may project from the first surface **42** and extend along a portion of the peripheral edge of the first surface **42**.

(101) The shape of the peripheral wall **46** is not limited to that in the embodiment. Since the peripheral wall **46** extends along the peripheral edge of the first surface **42**, the shape of the peripheral wall **46** may be changed in conformance with the shape of the first surface **42** as viewed in the third direction Z. The peripheral wall **46** may be omitted from the cover **13**.

(102) The cover **13** may include only one of the first reinforcement parts **58a** and **58b**.

Alternatively, the first reinforcement parts **58a** and **58b** may be omitted from the cover **13**. The cover **13** may include only one of the second reinforcement parts **59a** and **59b**. Alternatively, the second reinforcement parts **59a** and **59b** may be omitted from the cover **13**.

(103) In the present embodiment, the first guides **34a** and **34b** include the guide recesses **35a** and **35b**, respectively. The second guides **57a** and **57b** are fitted into the guide recesses **35a** and **35b**, respectively. Instead, the second guides **57a** and **57b** may include a guide recess into which the corresponding one of the first guides **34a** and **34b** is fitted.

(104) Of the first guides **34a** and **34b** and the second guides **57a** and **57b**, the cover **13** may include only the first guide **34a** and the second guide **57a**. Of the first guides **34a** and **34b** and the second guides **57a** and **57b**, the cover **13** may include only the first guide **34b** and the second guide **57b**. The first guides **34a** and **34b** and the second guides **57a** and **57b** may be omitted from the cover **13**.

(105) The length of the second reinforcement rib **47** projected from the second surface **43** may be shorter than the distance from the second surface **43** to the connection surface **23**.

(106) The cover **13** does not necessarily have to include the sealant **44d**. The sealing portion **44** may be omitted from the cover **13**.

(107) The shape of the cover **13** is not limited to that in the embodiment. The cover **13** may have any shape that includes at least one first reinforcement rib **45** and at least one second reinforcement rib **47**. In an example, the shape of each of the projection walls **51a** and **51b**, the fixing portions **52a** and **52b**, and the rib walls **56a** to **56d** may be changed. The cover **13** may further include a reinforcement rib that projects from the exposed surface **43a** and extends in the first direction X. In this case, the dimension of the reinforcement rib in the third direction Z is set so that the reinforcement rib does not contact the connection parts **21**.

(108) FIGS. **9** and **10** show a second modified example of a cover **13A** that includes a protrusion **71**. In FIGS. **9** and **10**, the same reference characters are given to those components that are the same as the corresponding components of the embodiment.

(109) As shown in FIGS. **9** and **10**, the cover **13A** includes a first end surface **72** and a second end surface **73** in the second direction Y. One of the first end surface **72** and the second end surface **73** includes an outward surface **74** extending in the first direction X. In an example, each of the end surface **72** and the end surface **73** includes the outward surface **74**. The outward surface **74** of the

end surface **72** is referred to as an outward surface **74a**. The outward surface **74** of the end surface **73** is referred to as an outward surface **74b**. In an example, the outward surfaces **74a** and **74b** are planar and orthogonal to the second direction Y. In an example, the outward surface **74a** is parallel to the outward surface **74b**.

(110) The cover **13A** includes a protrusion **71** protruding from the outward surface **74a** in the second direction Y. In an example, the protrusion **71** is located in a second end region of the cover **13A** in the first direction X. The protrusion **71** includes a distal end in the protrusion direction, that is, the protrusion **71** includes a distal end in the second direction Y. The distal end is located at an outer side of the cover **13A** from the sealant **44d** as viewed in the third direction Z.

(111) With this structure, when the cover **13A** is not coupled to the housing **12** and is placed on a surface **S1** of a ground, a floor, or the like, the cover **13A** may be located so that the outward surface **74a**, from which the protrusion **71** protrudes, faces the surface **S1**. FIG. **10** shows the cover **13A** that is located so that the outward surface **74a** faces the surface **S1** of a ground, a floor, or the like. In this case, the cover **13A** is located on the surface **S1** so that the outward surface **74a** is spaced apart from the surface **S1** by the protrusion **71**. That is, the cover **13A** is partially lifted by the protrusion **71** and is located on the surface **S1**. Thus, the sealing portion **44** is likely to be disposed at a position separated upward from the surface **S1**. This limits contact of the surface **S1** with the sealant **44d**, which is disposed on the outer surface of the sealing portion **44**. Accordingly, objects such as sand or dust on the surface **S1** are less likely to collect on the sealant **44d**. When the cover **13A** is coupled to the housing **12**, objects are less likely to be caught between the sealant **44d** and the housing **12**. Thus, the sealant **44d** is in satisfactory tight contact with the housing **12**. The sealant **44d** demonstrates a sufficient sealing property.

(112) The protrusion **71** is disposed on the outward surface **74a** but is not disposed on the outward surface **74b**. When a person couples the cover **13A** to the housing **12**, the protrusion **71** is used as a mark for determining the orientation of the cover **13A** with respect to the housing **12**. Thus, coupling of the cover **13A** to the housing **12** at an improper position is avoided. In addition, the task efficiency of coupling the cover **13A** to the housing **12** is improved.

(113) The cover **13A** may include at least one protrusion **71** protruding from the outward surface **74b** in a direction opposite to the second direction Y. In this case, in the cover **13A**, the outward surface **74a** may include at least one protrusion **71**. Alternatively, the protrusion **71** may be omitted from the outward surface **74a**.

(114) In the cover **13A** shown in FIGS. **9** and **10**, the fixing portion **52a** projects from the distal end of the projection wall **51a** in the first direction X. As viewed in the third direction Z, the fixing portion **52b** projects from the distal end of the projection wall **51b** in a direction intersecting the first direction X. In an example, as viewed in the third direction Z, the distal end of the fixing portion **52b** is located at a position separated from the distal end of the fixing portion **52a** in the second direction Y.

(115) As shown in FIG. **10**, as viewed in the third direction, the fixing hole **53a** in the fixing portion **52a** and the fixing hole **53b** in the fixing portion **52b** may be located at different positions in the second direction Y. In FIG. **10**, the single-dashed line refers to a center line **L1** that extends parallel to the first direction X through the center between the outward surface **74a** and the outward surface **74b** in the second direction Y as viewed in the third direction Z. In an example, as viewed in the third direction Z, the center of the fixing hole **53a** is located on the center line **L1**. In an example, as viewed in the third direction Z, the center of the fixing hole **53b** is located at a position separated from the center line **L1** in the second direction Y.

(116) As shown in FIGS. **9** and **10**, the cover **13A** may include fixing portion ribs **75a** and **75b** projecting from the fixing portions **52a** and **52b**, respectively.

(117) The fixing portion rib **75a** projects from the fixing portion **52a** in the third direction Z. In an example, the fixing portion rib **75a** includes edge ribs **76a** and **76b** and a joint portion **77a**. The edge rib **76a** extends along a first end of the fixing portion **52a** in the second direction Y. The edge

rib **76b** extends along a second end of the fixing portion **52a** in the second direction Y. Each of the edge ribs **76a** and **76b** includes a second end in the first direction X joined to the projection wall **51a**. Each of the edge ribs **76a** and **76b** projects from the fixing portion **52a** by a length that decreases as the projection wall **51a** becomes farther in the first direction X.

(118) The joint portion **77a** projects from the fixing portion **52a** in the third direction Z. The joint portion **77a** joins the edge rib **76a** and the edge rib **76b** on a first end surface of the fixing portion **52a** in the third direction Z. As viewed in the third direction Z, the joint portion **77a** is disposed between the projection wall **51a** and the fixing hole **53a**. In an example, as viewed in the third direction Z, the joint portion **77a** is arc-shaped. The shape of the joint portion **77a** as viewed in the third direction Z is not limited to the shape of an arc and may be any shape that joins the edge rib **76a** and the edge rib **76b**. In an example, as viewed in the third direction Z, the shape of the joint portion **77a** may be linear, bent, or any type of curve.

(119) The fixing portion rib **75b** projects from the fixing portion **52b** in the third direction Z. In an example, the fixing portion rib **75b** includes edge ribs **76c** and **76d** and a joint portion **77b**. The edge rib **76c** extends along a first end of the fixing portion **52b** in the second direction Y. The edge rib **76d** extends along a second end of the fixing portion **52b** in the second direction Y. Each of the edge ribs **76c** and **76d** includes a first end in the first direction X joined to the projection wall **51b**. Each of the edge ribs **76c** and **76d** projects from the fixing portion **52b** by a length that decreases as the projection wall **51b** becomes farther in a direction opposite to the first direction X.

(120) The joint portion **77b** projects from the fixing portion **52b** in the third direction Z. The joint portion **77b** joins the edge rib **76c** and the edge rib **76d** on a first end surface of the fixing portion **52b** in the third direction Z. As viewed in the third direction Z, the joint portion **77b** is disposed between the projection wall **51b** and the fixing hole **53b**. In an example, as viewed in the third direction Z, the joint portion **77b** is arc-shaped. The shape of the joint portion **77b** as viewed in the third direction Z is not limited to the shape of an arc and may be any shape that joins the edge rib **76c** and the edge rib **76d**. In an example, as viewed in the third direction Z, the shape of the joint portion **77b** may be linear, bent, or any type of curve.

(121) With this structure, the rigidity of the cover **13A** near the fixing portions **52a** and **52b** is increased by the fixing portion ribs **75a** and **75b**. Thus, when the cover **13A** receives an external force, deformation of the cover **13A** is further restricted.

(122) As described above, in the cover **13A** shown in FIGS. **9** and **10**, the fixing holes **53a** and **53b** are located at different positions in the second direction Y as viewed in the third direction Z. Thus, when the cover **13A** receives an external force that pushes the exposed surface **43a** toward the outside of the housing **12**, stress concentration is more likely to occur than in the cover **13A** in which the fixing holes **53a** and **53b** are aligned with each other in the second direction Y as viewed in the third direction Z. For example, in the cover **13A**, stress may concentrate on one of the fixing portion **52a** having the fixing hole **53a** and the fixing portion **52b** having the fixing hole **53b**. In this regard, the fixing portion ribs **75a** and increase the rigidity of the fixing portions **52a** and **52b** and part of the cover **13A** located near the fixing portions **52a** and **52b**. Thus, although the fixing holes **53a** and **53b** are located at different positions in the second direction Y as viewed in the third direction Z, when an external force is applied to the cover **13A**, the cover **13A** is less likely to deform.

(123) The cover **13A** may include only the fixing portion rib **75a** or may include only the fixing portion rib **75b**.

(124) The cover **13A** shown in FIGS. **9** and **10** is formed by injection molding. The cover **13A** includes a gate mark **78** on the first surface **42**. The gate mark **78** is formed because the gate is set to be located on the first surface **42** when the cover **13A** is formed by injection molding. As viewed in the third direction Z, the gate mark **78** is located at a position separated from the first reinforcement ribs **45** in the second direction Y. As viewed in the third direction Z, the gate mark **78** is not lined with the first reinforcement ribs **45** in the first direction X. In an example, as viewed in

the third direction Z, the gate mark **78** is located between two first reinforcement ribs **45**.

Alternatively, as viewed in the third direction Z, the gate mark **78** may be located between one of the two first reinforcement ribs **45** and part of the peripheral wall **46** that is located adjacent to the first reinforcement rib **45** in the second direction Y.

(125) Thus, the gate mark **78** is located avoiding the first reinforcement ribs **45**. This allows the first reinforcement ribs **45** to continuously extend in the first direction X from the first end of the first surface **42** in the first direction X to the second end of the first surface **42** in the first direction X. Thus, the rigidity of the cover **13A** is increased as compared to a structure in which each first reinforcement rib is separated into pieces in the first direction X.

(126) The housing **12** is not limited to the shape of the embodiment and may have any shape that includes the opening **11** and is configured to hold the terminals **14**.

(127) In the embodiment described above, the first direction X refers to the direction extending from the connection part **21a** toward the connection part **21c**. Alternatively, the first direction X may refer to a direction extending from the connection part **21c** toward the connection part **21a**. In the embodiment described above, the second direction Y orthogonally intersects the first direction X. Alternatively, the angle at which the second direction Y intersects the first direction X may be changed in accordance with the shape of the connection part **21**. The third direction Z may differ from that of the embodiment and may be any direction that orthogonally intersects the first direction X and the second direction Y.

(128) The present disclosure includes the following modes. To facilitate understanding, the reference signs of some of the components of the exemplary embodiment are provided with no intention to limit. In the following modes, some of the elements may be omitted or may be selected or extracted to be combined with each other. [Clause 1] A connector (**10**) according to a mode of the present disclosure, including: terminals (**14**) arranged next to each other, each of the terminals (**14**) including a connection part (**21**) configured to be electrically connected to a mate terminal (**9**); a connector housing (**12**) that holds the terminals (**14**), the connector housing (**12**) defining an accommodation cavity that accommodates the connection parts (**21**) arranged next to each other and an opening (**11**) connected to the accommodation cavity; and a cover (**13**) that closes the opening (**11**), in which the cover (**13**) may include a first surface (**42**) located at a side opposite to the connector housing (**12**), at least one first reinforcement rib (**45**) extending on the first surface (**42**) in a longitudinal direction of the cover (**13**), a second surface (**43**) covering the accommodation cavity, and at least one second reinforcement rib (**47**) extending on the second surface (**43**) along the connection parts (**21**). [Clause 2] In an embodiment, the second reinforcement rib (**47**) and the first reinforcement rib (**45**) may project in opposite directions. [Clause 3] In an embodiment, the second reinforcement rib (**47**) and the first reinforcement rib (**45**) may extend in different directions. [Clause 4] In an embodiment, the cover (**13**) may include a sealing portion (**44**) projecting from the second surface (**43**) and extending along the opening (**11**). [Clause 5] In an embodiment, the first surface (**42**) may be located at a side opposite to the second surface (**43**). [Clause 6] In an embodiment, the second reinforcement rib (**47**) may intersect the sealing portion (**44**). [Clause 7] In an embodiment, the cover (**13**) may include first reinforcement ribs (**45**) arranged parallel to each other. [Clause 8] In an embodiment, the cover (**13**) may include second reinforcement ribs (**47**) arranged parallel to each other. [Clause 9] In an embodiment, the second reinforcement ribs (**47**) may partition the accommodation cavity.

REFERENCE SIGNS LIST

(129) **9**) mate terminal **10**) connector **11**) opening **12**) connector housing (housing) **13**, **13A**) cover **14**) terminal **21**, **21a** to **21c**) connection part **22**) terminal main body **23**) connection surface **25**) electric wire **31**) housing main body **32**) insertion opening **33**) electric wire holder **34a**, **34b**) first guide **35b**) guide recess **41**) closing portion **42**) first surface **43**) second surface **43a**) exposed surface **43b**) non-facing surface **44**) sealing portion **44a**, **44b**) side wall **44c**) sealing groove **44d**) sealant **45**) first reinforcement rib **46**) peripheral wall **47**) second reinforcement rib **48**) joint rib **49**)

auxiliary rib 51a, 51b) projection wall 52a, 52b) fixing portion 53a, 53b) fixing hole 56a to 56d) rib wall 57a, 57b) second guide 58a, 58b) first reinforcement part 59a, 59b) second reinforcement part 61a, 61b) third reinforcement rib 71) protrusion 72, 73) end surface 74, 74a, 74b) outward surface 75a, 75b) fixing portion rib 76a to 76d) edge rib 77a, 77b) joint portion 78) gate mark D1, D2, D3) distance L1) center line S1) surface X) first direction Y) second direction Z) third direction

Claims

1. A connector comprising: a connector housing including an opening; a cover closing the opening; and terminals held by the connector housing, wherein each of the terminals includes a connection part configured to be electrically connected to one of a plurality of mate terminals, the mate terminals differing from one another, the connection parts are arranged in a line in the connector housing, the cover includes a first surface exposed outside the connector housing and a second surface located at a rear side of the first surface, the second surface includes an exposed surface exposed through the opening to an inner side of the connector housing, the cover further includes at least one first reinforcement rib projecting from the first surface and at least one second reinforcement rib projecting from the second surface, when a first direction refers to a direction in which the connection parts are arranged, a second direction refers to a direction intersecting the first direction and extending from a basal end to a distal end of the connection parts, and a third direction refers to a direction orthogonal to the first direction and the second direction, the first reinforcement rib extends in the first direction, and the second reinforcement rib extends in the second direction between adjacent ones of the connection parts as viewed in the third direction.
2. The connector according to claim 1, wherein the cover includes a sealing portion projecting from the second surface to surround the exposed surface as viewed in the third direction, the sealing portion is fitted into the opening, and at least one of the second reinforcement rib is greater than the sealing portion in a length of projection from the second surface.
3. The connector according to claim 1, wherein the cover includes a resin molding component, and as viewed in any of the second direction and the third direction, the second reinforcement rib is disposed between each two of the connection parts located adjacent to each other in the first direction.
4. The connector according to claim 1, wherein the second surface extends wider than the exposed surface in opposite directions along the first direction, the connector housing includes two first guides extending in the third direction from two ends of the connector housing located at opposite sides in the first direction, the cover includes two second guides projecting from two parts of the second surface located outward from the exposed surface in the first direction, and the two second guides are fitted to the two first guides using a recess and a projection to restrict misalignment of the connector housing with the cover in the first direction and the second direction.
5. The connector according to claim 4, wherein the cover includes two first reinforcement parts and/or two second reinforcement parts, the two first reinforcement parts extend on the second surface from the two second guides to a first end of the cover in the second direction, and the two second reinforcement parts extend on the second surface from the two second guides to a second end of the cover in the second direction.
6. The connector according to claim 1, wherein the cover includes a peripheral wall projecting from the first surface and extending along a peripheral edge of the first surface.
7. The connector according to claim 6, wherein the peripheral wall annularly extends along the peripheral edge of the first surface.
8. The connector according to claim 1, wherein the at least one second reinforcement rib includes second reinforcement ribs, and the cover includes at least one joint rib that joins ones of the second reinforcement ribs that are located adjacent to each other in the first direction.
9. The connector according to claim 1, wherein the cover includes two of the first reinforcement

rib.

10. The connector according to claim 1, wherein the cover includes two end surfaces located in the second direction, at least one of the two end surfaces includes an outward surface extending in the first direction, and the cover includes a sealing portion projecting from the second surface to surround the exposed surface as viewed in the third direction, the sealing portion is fitted into the opening, an annular sealant sealing a gap between the connector housing and an outer surface of the sealing portion that faces the connector housing, and a protrusion protruding from at least one of the outward surface in the second direction or a direction opposite to the second direction.
